RACE Regular Analysis Through Continuous Exercise

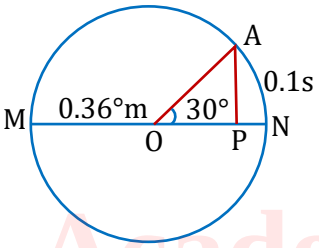
Simple Harmonic Motion (SHM) : RACE-01

1. A block of mass m attached to massless spring is performing oscillatory motion of amplitude 'A' on a frictionless horizontal plane. If half of the mass of the block breaks off when it is passing through its equilibrium point, the amplitude of oscillation for the remaining system become fA. The value of f is:

(A) $\frac{1}{2}$ (B) $\sqrt{2}$ (C) 1 (D) $\frac{1}{\sqrt{2}}$

PSH219

2. The point A moves with a uniform speed along the circumference of a circle of radius 0.36 m and covers 30° in 0.1 s. The perpendicular projection 'P' from 'A' on the diameter MN represents the simple harmonic motion of 'P'. The restoration force per unit mass when P touches M will be :



(A) 100 N (B) 0.49 N (C) 50 N (D) 9.87 N

PSH220

3. A particle executes S.H.M. with amplitude 'a' and time period V. The displacement of the particle when its speed is half of maximum speed is $\frac{\sqrt{x}a}{2}$. The value of x is .....

PSH221

4. A particle performs simple harmonic motion with a period of 2 second. The time taken by the particle to cover a displacement equal to half of its amplitude from the mean position is $\frac{1}{a}s$. The value of 'a' to the nearest integer is _____.

PSH222

5. A particle executes simple harmonic motion represented by displacement function as $x(t) = A \sin(\omega t + \phi)$. If the position and velocity of the particle at t = 0 s are 2 cm and 2ω $cm\ s^{-1}$ respectively, then its amplitude is $x\sqrt{2}cm$ where the value of x is _____.

PSH223

6. The equation of a particle executing simple harmonic motion is given by $x = \sin\pi\left(t + \frac{1}{3}\right)m$.

At t = 1s, the speed of particle will be (Given : π = 3.14)

(A) 0 $cm\ s^{-1}$ (B) 157 $cm\ s^{-1}$ (C) 272 $cm\ s^{-1}$ (D) 314 $cm\ s^{-1}$

PSH224

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7. The displacement of simple harmonic oscillator after 3 seconds starting from its mean position is equal to half of its amplitude. The time period of harmonic motion is :

(A) $6s$ (B) $8s$ (C) $12s$ (D) $36s$

PSH225

8. A particle executes simple harmonic motion. Its amplitude is 8 cm and time period is $6s$. The time it will take to travel from its position of maximum displacement to the point corresponding to half of its amplitude, is _____s.

PSH226

9. If two similar springs each of spring constant K are joined in series, the new spring constant and time period would be changed by a factor:

(A) $\frac{1}{2},\sqrt{2}$ (B) $\frac{1}{4},\sqrt{2}$ (C) $\frac{1}{4},2\sqrt{2}$ (D) $\frac{1}{2},2\sqrt{2}$

PSH227

10. Two particles A and B of equal masses are suspended from two massless springs of spring constants K and $4K$ respectively. If the maximum velocities during oscillations are equal, the ratio of the amplitude of A and B is

(A) $\sqrt{2}$ (B) 3 (C) 1 (D) 2

PSH228

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	D	3	6	2	B	D	1	A	D

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RACE Regular Analysis Through Continuous Exercise

Simple Harmonic Motion (SHM) : RACE-02

1. A small block is attached to an ideal spring and moving in SHM on a horizontal frictionless surface. The amplitude of the motion is 0.12 m. The maximum speed of the block is 3.6 m/s. What is the maximum magnitude of the acceleration of the block?

PSH229

2. When a particle executes SHM, the nature of graphical representation of velocity as a function of displacement is :

(A) circular (B) elliptical (C) parabolic (D) straight line

PSH230

3. A particle is executing Simple Harmonic Motion (SHM) of amplitude A, along the x-axis, about $x = 0$. When its Potential Energy (PE) equals Kinetic Energy (KE), the position of the particle will be:

(A) $\frac{A}{2}$ (B) $\frac{A}{2\sqrt{2}}$ (C) $\frac{A}{\sqrt{2}}$ (D) A

PSH231

4. A particle of mass 40 g executes a simple harmonic motion of amplitude 2.0 cm. If the time period is 0.20 s, find the total mechanical energy of the system.

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PSH232

5. In a simple harmonic oscillation, what fraction of total mechanical energy is in the form of kinetic energy, when the particle is midway between mean and extreme position.

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

PSH233

6. An object of mass 0.5 kg is executing simple harmonic motion. Its amplitude is 5 cm and time period (T) is 0.2s. What will be the potential energy of the object at an instant $t=\frac{T}{4}$ starting from mean position. Assume that the initial phase of the oscillation is zero.

(A) 0.62 J (B) 6.2×10^{-3} J (C) 1.2×10^{3} J (D) 6.2×10^{3} J

PSH234

7. For a body executing S.H.M. :

(a) Potential energy is always equal to its K.E.

(b) Average potential and kinetic energy over any given time interval are always equal.

(c) Sum of the kinetic and potential energy at any point of time is constant.

(d) Average K.E. in one time period is equal to average potential energy in one time period.

Choose the most appropriate option from the options given below :

(A) (c) and (d) (B) Only (c) (C) (b) and (c) (D) only (b)

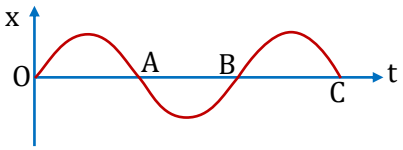
PSH235

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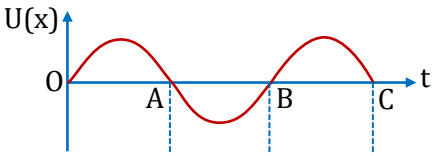
8. A body is performing simple harmonic with an amplitude of 10 *cm*. The velocity of the body was tripled by air Jet when it is at 5 *cm* from its mean position. The new amplitude of vibration is $\sqrt{x}$ cm. The value of *x* is_____.

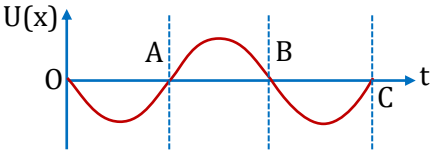
PSH236

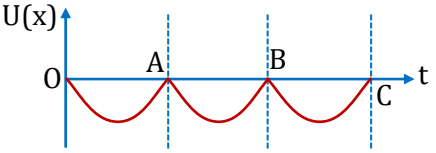
9. The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure.

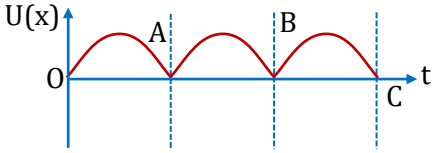


The potential energy $U(x)$ versus time (t) plot of the particle is correctly shown in figure :

(A)


(B)


(C)


(D)


PSH237

10. You are watching an object that is moving in SHM. When the object is displaced 0.600 *m* to the right of its equilibrium position, it has a velocity of 2.20 *m/s* to the right and an acceleration of 8.40 m/s^2 to the left. How much farther from this point will the object move before it stops immediately and then starts to move back to the left?

PSH238

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ANSWER KEY

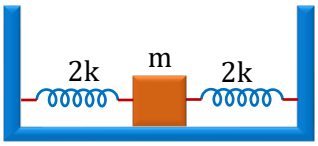
Question	1	2	3	4	5	6	7	8	9	10
Answer	108 m/s^2	B	C	7.9×10^{-3} *J*	B	A	A	700	D	0.24 *m* right

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Regular Analysis Through Continuous Exercise

Simple Harmonic Motion (SHM) : RACE-03

1. Two identical springs of spring constant '2k' are attached to a block of mass m and to fixed support (see figure). When the mass is displaced from equilibrium position on either side, it executes simple harmonic motion. The time period of oscillations of this system is :

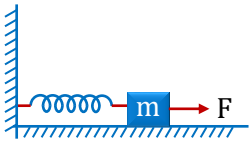


(A) $2\pi\sqrt{\frac{m}{k}}$ (B) $\pi\sqrt{\frac{m}{2k}}$ (C) $2\pi\sqrt{\frac{m}{2k}}$ (D) $\pi\sqrt{\frac{m}{k}}$

PSH239

2. A block of mass m, lying on a smooth horizontal surface, is attached to a spring (of negligible mass) of spring constant k. The other end of the spring is fixed, as shown in the figure. The block is initially at rest in its equilibrium position. If now the block is pulled with a constant force F, the maximum speed of the block is :

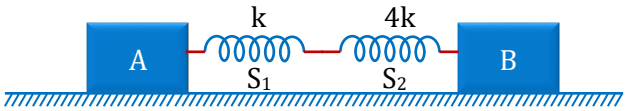
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(A) $\frac{\pi F}{\sqrt{mk}}$ (B) $\frac{2F}{\sqrt{mk}}$ (C) $\frac{F}{\sqrt{mk}}$ (D) $\frac{F}{\pi\sqrt{mk}}$

PSH240

3. In the reported figure, two bodies A and B of masses 200 g and 800 g are attached with the system of springs. Springs are kept in a stretched position with some extension when the system is released. The horizontal surface is assumed to be frictionless. The angular frequency will be _____ rad/s when $k = 20\ N/m$.



PSH241

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4. As per given figures, two springs of spring constants K and $2K$ are connected to mass m. If the period of oscillation in figure (a) is $3s$, then the period of oscillation in figure (b) will be $\sqrt{x}s$. The value of x is_____.

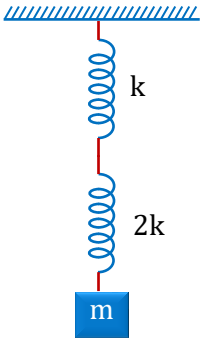


Figure (a)

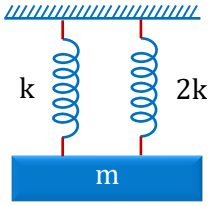


Figure (b)

PSH242

5. The motion of a simple pendulum executing S.H.M. is represented by following equation. $Y = A \sin(\pi t + \phi)$, where time is measured in second. The length of pendulum is :

(A) 97.23 cm (B) 25.3 cm (C) 99.4 cm (D) 406.1 cm

PSH243

6. Time period of a simple pendulum in a stationary lift is 'T'. If the lift accelerates with $\frac{g}{6}$ vertically upwards then the time period will be : (Where g = Acceleration due to gravity)

(A) $\sqrt{\frac{6}{5}}T$ (B) $\sqrt{\frac{5}{6}}T$ (C) $\sqrt{\frac{6}{7}}T$ (D) $\sqrt{\frac{7}{6}}T$

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PSH244

7. If the time period of a two meter long simple pendulum is 2s, the acceleration due to gravity at the place where pendulum is executing S.H.M. is:

(A) $\pi^2\, ms^{-2}$ (B) 9.8 ms^{-2} (C) $2\pi^2\, ms^{-2}$ (D) 16 m/s^2

PSH245

8. Given below are two statements :

Statement I : A second's pendulum has a time period of 1 second.

Statement II : It takes precisely one second to move between the two extreme positions.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false.

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true.

PSH246

9. T_0 is the time period of a simple pendulum at a place. If the length of the pendulum is reduced to $\frac{1}{16}$ times of its initial value, the modified time period is :

(A) T_0 (B) $8\pi T_0$ (C) $4T_0$ (D) $\frac{1}{4}T_0$

PSH247

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10. A bob of mass 'm' suspended by a thread of length l undergoes simple harmonic oscillations with time period T. If the bob is immersed in a liquid that has density $\frac{1}{4}$ times that of the bob and the length of the thread is increased by $1/3^{rd}$ of the original length, then the time period of the simple harmonic oscillations will be :-

(A) T (B) $\frac{3}{2}T$ (C) $\frac{3}{4}T$ (D) $\frac{4}{3}T$

PSH248

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	10	2	C	C	C	B	D	D

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RACE
Regular Analysis Through Continuous Exercise

Simple Harmonic Motion (SHM) : RACE-04

1. A particle is subjected to two simple harmonic motions $x_1 = A \sin \omega t$ and $x_2 = A \sin(\omega t + \pi/3)$. Find the maximum speed of the particle.

(A) $\sqrt{2}\, A\omega$ (B) $A\omega$ (C) $\sqrt{3}\, A\omega$ (D) $3\, A\omega$

PSH249

2. A particle is subjected to two simple harmonic motions in the same direction having equal amplitudes and equal frequency. If the resultant amplitude is equal to the amplitude of the individual motions, find the phase difference between the individual motions.

(A) $\frac{\pi}{2}$ (B) π (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{3}$

PSH250

3. The displacement y of a particle executing periodic motion is given by $y = 4\cos^2(t/2)\sin(1000t)$. This expression may be considered to be a result of the superposition of _____ independent harmonic motions.

(A) Two (B) Three (C) Four (D) Five

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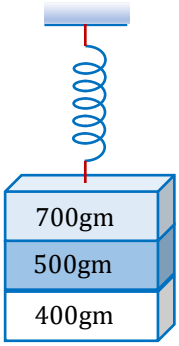
PSH251

4. A simple pendulum has some time period T. What will be the percentage change in its time period if its amplitude is decreased by 5% ?

(A) 6% (B) 3% (C) 1.5% (D) 0%

PSH252

5. Three masses $700g$, $500g$, and $400g$ are suspended at the end of a spring a shown and are in equilibrium. When the $700g$ mass is removed, the system oscillates with a period of 3 seconds, when the $500g$ mass is also removed, it will oscillate with a period of :-

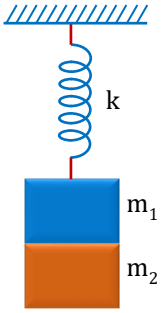


(A) $1s$ (B) $2s$ (C) $3s$ (D) $\sqrt{\frac{12}{5}}\, s$

PSH253

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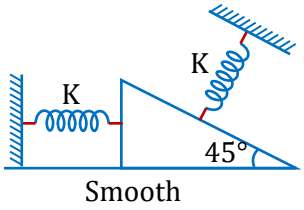
6. Two masses m_1 and m_2 are suspended together by a massless spring of constant k. When the masses are in equilibrium, m_1 is removed without disturbing the system; the amplitude of vibration is :-



(A) $m_1 g/k$ (B) $m_2 g/k$ (C) $\frac{(m_1+m_2)g}{k}$ (D) $\frac{(m_2-m_1)g}{k}$

PSH254

7. For the given situation, if wedge is slightly displaced and released then find out the time period of oscillation (initially springs are in natural length) :

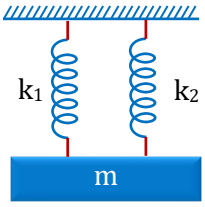


(A) $T = 2\pi\sqrt{\frac{m}{K}}$ (B) $T = 2\pi\sqrt{\frac{m}{2K}}$ (C) $T = 2\pi\sqrt{\frac{2m}{3K}}$ (D) None of these

PSH255

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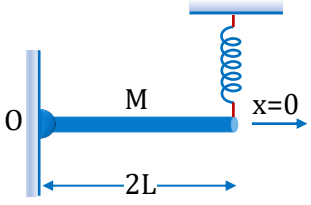
8. A mass is suspended separately by two different springs in successive order then time periods is t_1 and t_2 respectively. It is connected by both springs as shown in figure, then time period is t, the correct relation is



(A) $t_0^2 = t_1^2 + t_2^2$ (B) $t_0^{-2} = t_1^{-2} + t_2^{-2}$ (C) $t_0^{-1} = t_1^{-1} + t_2^{-1}$ (D) $t_0 = t_1 + t_2$

PSH256

9. A uniform pole of length $\ell = 2L$ is laid on smooth horizontal table as shown in figure. The mass of pole is M and it is connected to a frictionless axis at O. A spring with force constant k is connected to the other end. The pole is displaced by a small angle θ_0 from equilibrium position and released such that it performs small oscillations. Then

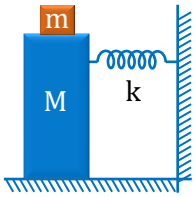


(A) $\omega_0 = \sqrt{\frac{M}{3k}}$ (B) $\omega_0 = \sqrt{\frac{k}{3M}}$ (C) $\omega_0 = \sqrt{\frac{3k}{M}}$ (D) $\omega_0 = \sqrt{\frac{k}{2M}}$

PSH257

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10. In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k. The mass oscillates on a frictionless surface with time period T and amplitude A. When the mass is in equilibrium position, as shown in the figure, another mass m is gently fixed upon it. The new amplitude of oscillation will be :



(A) $A\sqrt{\frac{M-m}{M}}$ (B) $A\sqrt{\frac{M}{M+m}}$ (C) $A\sqrt{\frac{M+m}{M}}$ (D) $A\sqrt{\frac{M}{M-m}}$

PSH258

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	B	D	B	A	C	B	B	B

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